

Clinical Risk Assessment Report: Patient 89

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 12.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.56x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.9%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: -0.34 [Bottom 14%] (+3.4%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.52 [Bottom 18%] (+1.4%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: -0.78 [Bottom 19%] (-3.9%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Patient Age (Age)

Value: 74.0 [Top 13%] (-1.8%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

Value: 0.1 [Avg (41th %ile)] (-1.6%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)

Value: -0.42 [Bottom <1%] (-0.8%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.
- Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET)

Value: -0.33 [Avg (30th %ile)] (-0.7%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.
- Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT)

Value: 0.42 [Avg (57th %ile)] (-0.5%)

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)

